

Robustness across conditions

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Inspecting failures and anomalies

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Robustness across conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Method passes9/10 on current sources and5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Robustness is bounded by the tested variations and independent evidence policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled robustness across conditions evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of robustness across conditions. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Robustness across conditions

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Fixed model across conditions

correct cases out of 8 per condition



Bar length shows the supplied value; the marked baseline is zero.

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.



2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined $10/16=62.5\%$.

Explain the changed result using the matching lesson.

Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The large observed gap motivates investigation rather than claiming condition independence.

1. Name both slices
2. Compute rates
3. Bound robustness

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound robustness The large observed gap motivates investigation rather than claiming condition independence. Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Practice 2: explain the deciding evidence

Changed-case practice: Report only the combined $10/16=62.5\%$. Which conclusion is supported by the supplied facts?

1. The arithmetic is correct, but the condition gap becomes hidden.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Repair a missing building block

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the mechanism and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the mechanism and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the output so another team can identify the exact run?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the mechanism and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

